

# Beyond Text: Evaluating Emoji Contributions to Social Media Engagement Prediction

## 0. Colab links

All analyses were conducted using two Google Colab notebooks.

- The notebook for models trained with class-weight adjustments is available at:  
[https://colab.research.google.com/drive/115yCPCfCbux8F9\\_W7zzWY1WHVkyWcfbs#scrollTo=Rt6kcDK\\_FQMT](https://colab.research.google.com/drive/115yCPCfCbux8F9_W7zzWY1WHVkyWcfbs#scrollTo=Rt6kcDK_FQMT)
- The notebook for models trained without class-weight adjustments is available at:  
[https://colab.research.google.com/drive/1v5ZgF93aMQiC\\_XRZFbYbtV-ieh1Q3Jir](https://colab.research.google.com/drive/1v5ZgF93aMQiC_XRZFbYbtV-ieh1Q3Jir)
- The GitHub repository is available at:  
[https://github.com/tzuhsi711/MIT\\_15773\\_Final\\_Project\\_Mar2026](https://github.com/tzuhsi711/MIT_15773_Final_Project_Mar2026)

## 1. Introduction

In the modern data-driven economy, social media has become a central platform for customer interaction, brand visibility, and real-time insight into consumer sentiment. Higher levels of consumer–brand engagement on these platforms are closely linked to improved conversion and purchase intention [1], making engagement a key driver of retail performance, particularly for small and medium enterprises (SMEs) [2]. Research further indicates that content characteristics such as readability and depth are positively associated with higher engagement and audience awareness [2]. Therefore, identifying which features of text-based social media posts drive engagement is essential for businesses.

In this study, we developed natural language processing (NLP) models to predict the engagement level of text-based posts across different social media platforms, including Facebook, Twitter, and Instagram, in order to generate actionable insights for customer experience management, brand monitoring, and market intelligence. Specifically, we constructed two models: one using text-only inputs and another using text together with textualized emojis. By comparing these two configurations, we aim to evaluate whether emoji-derived information improves the model’s ability to predict high-engagement posts.

## 2. Method

### 2.1 Data Preparation

We conducted our analysis using the SocialBuzz Sentiment Analytics dataset (Kaggle, 2026), which contains 732 text-based social media posts collected between 2010 and 2023 and annotated with engagement metrics. For each post, we focused on the textual content and the number of likes, treating the latter as a proxy for audience engagement. The dataset was randomly partitioned into training (80%, 586 posts), validation (10%, 73 posts), and test (10%, 73 posts) sets using a fixed random seed to ensure reproducibility. Each input was encoded into a maximum of 41 tokens and converted to IDs and attention masks with the RoBERTa tokenizer.

To frame engagement prediction as a binary classification problem, we transformed the continuous “Likes” variable into categorical labels. Specifically, we computed the 70th percentile of the likes distribution and labeled posts with likes greater than or equal to this threshold as high engagement, while posts below the threshold were labeled as low engagement. This labeling strategy allowed us to

focus on identifying posts that substantially outperform the majority in terms of audience response while still preserving a sufficient number of examples in both classes for model training.

Prior to modeling, we applied a consistent preprocessing pipeline to all posts. User mentions were anonymized by replacing them with the placeholder token “@user”, and URLs were normalized to the token “http” to reduce noise while preserving the structural signal of link presence. Because our primary goal was to evaluate the contribution of emojis to engagement prediction, we constructed two parallel versions of the dataset. In the text-only version, emojis were removed entirely from the posts. In the text-plus-emoji version, we used the Python Demoji library (version 1.1.0) to detect and convert emojis into their corresponding textual descriptions so that they could be processed as additional tokens by the language model. The resulting corpora were then tokenized using the tokenizer associated with the chosen pre-trained model and truncated or padded to a fixed maximum sequence length prior to training.

## **2.2 Model Architecture**

This study uses a pre-trained Twitter-RoBERTa-base model (cardiffnlp/twitter-roberta-base-sentiment-latest) as the backbone architecture. This model is based on the RoBERTa-base architecture and was pre-trained on approximately 124 million tweets before being fine-tuned for sentiment analysis on the TweetEval benchmark [3]. Because our training dataset is relatively small, containing 586 posts, one possible strategy would be to freeze the backbone and train only a shallow classifier. However, because our task of predicting high versus low engagement differs from sentiment classification, we fine-tuned the entire model end-to-end so that the pre-trained representations could adapt more effectively to engagement-related signals present in our dataset.

The resulting model consists of three components. The first component is the pre-trained Twitter-RoBERTa backbone, which produces contextualized token and sequence representations. The second component is a fully connected dense layer with ReLU activation, which maps the pooled sequence representation into a lower-dimensional hidden space. The third component is a final output layer with a single neuron and sigmoid activation, producing the probability that a given post belongs to the high-engagement class in the binary classification task. Using this architecture, we trained two separate models. The first model is a text-only model (TO), which receives posts with emojis removed. The second model is a text-plus-emoji model (TE), which receives posts with emojis converted into textual descriptions as additional tokens.

## **2.3 Training and Fine-Tuning**

Our main experiments focused on two weighted configurations: a text-only model with class-frequency-based class weights (TO-Weighted) and a text-plus-emoji model with class weights (TE-Weighted). Because the high-engagement class is underrepresented in our dataset, we incorporated class weights during fine-tuning to mitigate the effect of class imbalance. The class weight for each class  $c$  was computed as:

$$w_c = \frac{N}{K \cdot N_c}$$

where  $N$  represents the total number of training examples,  $K$  is the number of classes, and  $N_c$  is the number of training examples in class  $c$ . This procedure assigns a larger weight to the minority high-engagement class and a smaller weight to the majority low-engagement class. In our dataset, this resulted in a weight of 0.59 for the low-engagement class and 3.25 for the high-engagement class.

These weights were passed to the training routine so that the binary cross-entropy loss for each example was multiplied by the corresponding class weight.

All models were trained using binary cross-entropy loss, treating engagement prediction as a binary classification task between high- and low-engagement posts. We used a fixed batch size of 32, which provided a practical compromise between computational efficiency and stable gradient estimates on our hardware. Each model was trained for five epochs, as preliminary experiments indicated that validation performance plateaued or began to degrade beyond this point, suggesting the onset of overfitting on our relatively small dataset [4].

In addition to the weighted models, we also trained two baseline models without class weights, referred to as TO-Unweighted and TE-Unweighted. These models used identical architectures and hyperparameters but with all class weights set to one. These baselines allow us to directly compare the effect of class weighting on model performance under class imbalance.

To identify effective hyperparameter configurations for both weighted models, we used Keras Tuner (version 1.4.8) to search over the size of the dense layer (32, 64, 128), the dropout rate (0.1, 0.2, 0.3), and the learning rate (0.001, 0.0001, 0.00001). In both weighted configurations, we used the Adam optimizer because it provides a good balance between training speed and optimization stability for transformer-based models [5].

## **2.4 Evaluation Metrics**

We evaluated model performance on the held-out test set using standard metrics for binary classification, treating the high-engagement class as the positive class. We report overall accuracy, precision, recall, and F1-score. Accuracy measures the proportion of correctly classified posts across all predictions. Precision for the high-engagement class is defined as the fraction of posts predicted as high engagement that are truly high engagement, while recall measures the fraction of true high-engagement posts correctly identified by the model. The F1-score represents the harmonic mean of precision and recall, providing a balanced measure that is particularly informative when class distributions are imbalanced.

$$Precision = \frac{True\ positives}{True\ positives + False\ positives}$$

$$Recall = \frac{True\ positives}{True\ positives + False\ negatives}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

In addition, we report the area under the receiver operating characteristic curve (ROC-AUC), which quantifies the model's ability to distinguish between high- and low-engagement posts across different decision thresholds. To further analyze model behavior, we also compute confusion matrices on the test set. The confusion matrix summarizes the number of true positives, false positives, true negatives, and false negatives for the high-engagement class, allowing us to examine the types of classification errors made by each model.

To further interpret the predictions of the trained models, we applied SHAP (SHapley Additive exPlanations) to analyze token-level contributions to the predicted probabilities. SHAP assigns an importance value to each token in the input text, indicating whether the token increases or decreases

the predicted probability relative to a baseline prediction. This approach allows us to visually examine how specific words and emoji-derived tokens influence the model's engagement predictions.

### 3. Results

#### 3.1 Optimal hyperparameter configuration

The best-performing TO-Weighted model used a dense layer with 128 neurons, a dropout rate of 0.1, and a learning rate of 0.00001. The model consists of a Twitter-RoBERTa backbone that produces a 768-dimensional pooled representation, followed by a fully connected dense layer and a final sigmoid output layer. The classifier head contains 124,744,193 trainable parameters, including 124,645,632 parameters from the backbone, 98,432 parameters in the dense layer and 129 parameters in the output layer.

The best-performing TE-Weighted model used a smaller hidden layer with 32 neurons, a dropout rate of 0.3, and a learning rate of 0.001. Similar to the TO model, the Twitter-RoBERTa backbone remained frozen and only the classifier layers were trained. The classifier head contains 124,670,273 trainable parameters, including 124,645,632 parameters from the backbone, 24,608 parameters in the dense layer and 33 parameters in the output layer.

In Keras 3.x, parameters from external TensorFlow models may not be fully tracked, so the backbone may appear with 0 parameters in the model summary. The backbone weights are still included in the reported counts above. The Twitter-RoBERTa backbone follows the RoBERTa-base architecture, which contains 124,645,632 parameters.

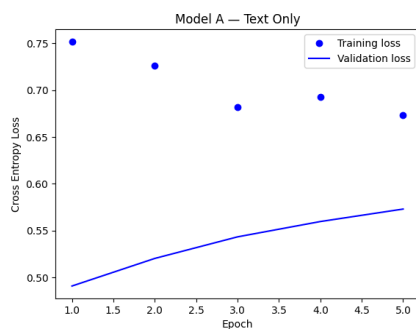
#### 3.2 Training and validation loss

For the TO-weighted model, the training loss decreased steadily across the five epochs and reached approximately 0.674 by the final epoch (Fig. 1a). In contrast, the validation loss increased gradually during training while remaining lower than the training loss.

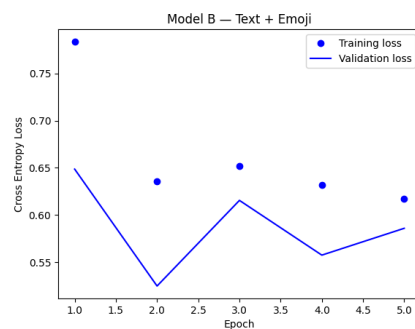
For the TE-weighted model, the training loss also decreased across the five epochs and reached approximately 0.638 by the final epoch (Fig. 1b). The validation loss fluctuated during training and remained lower than the training loss.

**Fig. 1 Training and validation loss for TO-Weighted and TE-Weighted models.**

**a.**



**b.**



(a) TO-Weighted model (b) TE-Weighted model

### 3.2 Overall Performance

We compared the quantitative performance of all four model configurations, including two weighted models and two unweighted models, on the held-out test set, treating the high-engagement class as the positive class. The weighted models substantially improved performance on the high-engagement class compared with their unweighted counterparts (Table 1). The TO-Weighted model achieved an accuracy of 0.61 and an F1-score of 0.29 for the high-engagement class, with precision of 0.20 and recall of 0.55. The TE-Weighted model showed a similar pattern, with a slightly lower accuracy of 0.57 and F1-score of 0.27, while maintaining the same recall of 0.55. Both weighted models also achieved higher ROC-AUC values, with 0.66 for TO-Weighted and 0.61 for TE-Weighted, indicating improved discrimination between high- and low-engagement posts across decision thresholds.

In contrast, the unweighted baseline models, TO-Unweighted and TE-Unweighted, achieved relatively high overall accuracy of 0.85 but had precision, recall, and F1-scores of zero for the high-engagement class. This result indicated that these models never predicted the positive class on the test set. This behavior reflected the underlying class imbalance in the dataset. By always predicting the majority low-engagement class, the unweighted models achieved high accuracy while completely failing to identify high-engagement posts. These findings demonstrated that incorporating class weights sacrificed some overall accuracy but was essential for recovering high-engagement posts, which was the primary objective of the modeling task.

Table 1: Test performance of all model configurations

| Model         | Accuracy | Precision | Recall | F <sub>1</sub> | ROC-AUC |
|---------------|----------|-----------|--------|----------------|---------|
| TO-Weighted   | 0.61     | 0.20      | 0.55   | 0.29           | 0.66    |
| TE-Weighted   | 0.57     | 0.18      | 0.55   | 0.27           | 0.61    |
| TO-Unweighted | 0.85     | 0.00      | 0.00   | 0.00           | 0.56    |
| TE-Unweighted | 0.85     | 0.00      | 0.00   | 0.00           | 0.47    |

### 3.3 Confusion Matrix

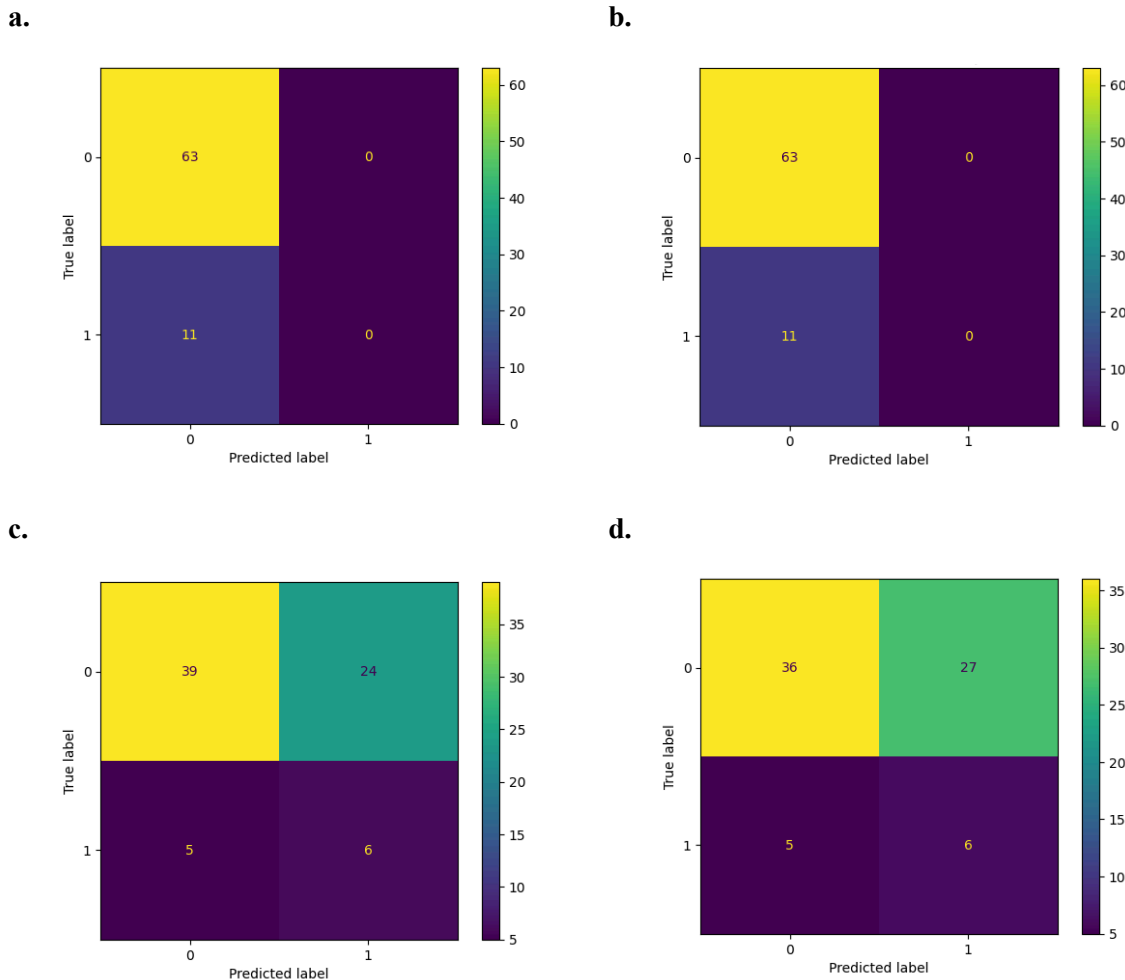
The confusion matrices provide a more detailed view of how the models classified individual examples across the two engagement classes (Fig. 2). Consistent with the quantitative results, the unweighted models predominantly predicted the majority low-engagement class, resulting in no correctly identified high-engagement posts (Fig. 2a–b). This pattern reflects the strong class imbalance in the dataset and highlights the limitation of relying solely on overall accuracy when evaluating model performance.

In contrast, the weighted models produced a comparatively more balanced distribution of predictions by identifying a subset of high-engagement posts while also introducing some false positives (Fig. 2c–d). This shift indicates that class weighting adjusted the decision boundary toward the minority class, allowing the models to recover high-engagement examples that would otherwise be missed.

A closer comparison of the two weighted models reveals a small trade-off in classification behavior. For the low-engagement class, the TO-Weighted model correctly classified 39 examples and misclassified 24 examples as high engagement (Fig. 2c), whereas the TE-Weighted model correctly classified 36 examples and misclassified 27 examples (Fig. 2d). Thus, the TO-Weighted model produced slightly more true negatives and fewer false positives than the TE-Weighted model. This pattern is consistent with the slightly higher accuracy and F1-score observed for the TO-Weighted

configuration and further suggests that incorporating emoji-derived tokens did not provide a clear performance advantage over the text-only model in this dataset.

**Fig. 2 Confusion matrices of weighted and unweighted models.**



(a) TO-Unweighted (b) TE-Unweighted (c) TO-Weighted (d) TE-Weighted

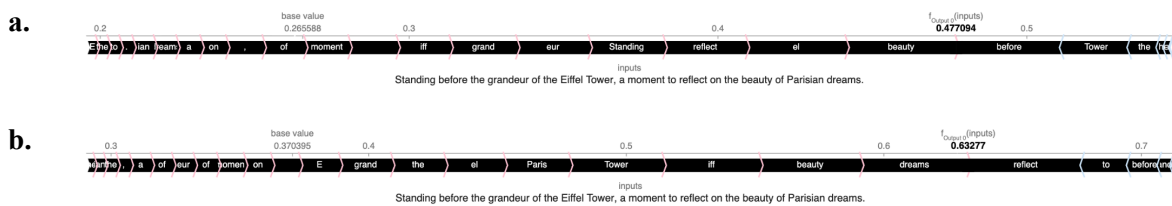
### 3.4 SHAP Feature Extraction

To better understand how the trained models form predictions, we applied SHAP (SHapley Additive exPlanations) to analyze token-level contributions for individual text inputs. SHAP assigns an importance value to each token in the input text, indicating whether the token increases or decreases the predicted probability relative to a baseline prediction.

For the TO-Weighted model, the baseline prediction probability was approximately 0.27, and the final predicted probability increased to approximately 0.48 after incorporating token-level contribution (Fig. 3a). Tokens such as “beauty”, “reflect”, and “grand” contributed positively and increased the predicted probability. In contrast, tokens such as “before”, “Tower”, and “the” contributed negatively and reduced the predicted probability. These results suggest that descriptive tokens associated with positive sentiment tend to increase the model’s predicted probability, while contextual tokens related to location or sentence structure have smaller or negative contributions.

For the TE-Weighted model, the baseline prediction probability was approximately 0.37, and the final predicted probability increased to approximately 0.63 after incorporating token-level contributions (Fig. 3b). Similar to the text-only model, tokens such as “beauty”, “dreams”, and “grand” contributed positively and increased the probability of the positive class. However, some tokens showed different contribution patterns between the two models. For example, “reflect”, which contributed positively in the TO-Weighted model, contributed negatively in the TE-Weighted model. Likewise, “Tower”, which had a negative contribution in the TO-Weighted model, contributed positively in the TE-Weighted model. In contrast, tokens such as “to”, “before”, and “and” contributed negatively and reduced the predicted probability.

**Fig. 3 SHAP explanation of token contributions for representative input text.**



(a) TO-Weighted model (b) TE-Weighted model

Overall, tokens such as “beauty” and “grand” consistently contributed positively across both models, suggesting that descriptive sentiment-related words play a dominant role in the prediction of engagement. In contrast, several contextual tokens showed model-specific contributions, indicating that the inclusion of emoji-derived tokens may alter how certain words influence the model’s predictions. Despite these differences, the most influential token-level contributions were still driven by descriptive textual features rather than emoji-derived information. For the example input, the TE-Weighted model produced a higher final predicted probability than the TO-Weighted model, but this increase was largely associated with textual tokens rather than emoji-related features.

#### 4. Discussion

The primary objective of this study was to evaluate whether incorporating emoji-derived information improves the ability of NLP models to predict engagement levels in text-based social media posts. Overall, the results suggest that including emoji-derived tokens does not noticeably improve model performance. With the exception of recall, the TO-Weighted model achieved higher overall accuracy, precision, F1-score, and ROC-AUC than the TE-Weighted model, indicating that the semantic information contained in textual content may already capture much of the signal necessary for predicting engagement in this dataset.

Examination of the training and validation loss curves further provides insight into the training behavior of the models. For the TO-Weighted model, the divergence between decreasing training loss and increasing validation loss suggests the onset of mild overfitting during the later training epochs. In contrast, the TE-Weighted model exhibited fluctuating validation loss rather than a consistent increase, indicating less systematic overfitting but potentially unstable generalization across epochs. Notably, the validation loss remained lower than the training loss for both models, which may partly reflect the use of dropout during training. Because dropout randomly deactivates neurons during

training but not during evaluation, the training objective becomes more difficult, leading to slightly higher training loss compared with validation loss.

To further examine the effect of class imbalance on model performance, we also compared the weighted models with baseline models trained without class-weight adjustments, including the TO-Unweighted and TE-Unweighted configurations. Although the unweighted models achieved much higher overall accuracy, they failed to identify any high-engagement posts, resulting in zero precision, recall, and F1-score for the positive class. This result demonstrates that overall accuracy alone may be misleading when evaluating model performance under class imbalance and highlights the importance of applying class-weight adjustments when predicting engagement in imbalanced datasets.

The SHAP analysis further illustrates how individual tokens influence model predictions. Token-level explanations indicate that both models rely primarily on descriptive and sentiment-related textual tokens when predicting engagement. Words associated with positive sentiment, such as “beauty” and “grand” tended to increase the predicted probability of high engagement, whereas contextual tokens related to sentence structure such as “before” generally contributed less strongly or negatively. Although the model incorporating emoji-derived tokens produced a slightly higher predicted probability for the example input, the most influential contributions were still driven by descriptive textual tokens rather than emoji-derived features. This pattern suggests that textual content plays a larger role than emoji-derived information in driving engagement predictions in this dataset.

## 5. Conclusion and Future Directions

In this study, we developed NLP-based models to predict engagement levels for text-based social media posts and evaluated whether incorporating emoji-derived information improves predicted performance. The text-only weighted model slightly outperformed text-plus-emoji model, suggesting that textual features alone capture much of the signal associated with engagement.

We also found that class imbalance is critical when predicting engagement outcomes. While unweighted models achieved higher overall accuracy, they failed to identify high-engagement posts, highlighting a trade-off between accuracy and minority-class detection.

However, several limitations should be considered when interpreting these findings. First, the dataset only included posts collected from Facebook, Twitter, and Instagram, which may limit the generalizability of the model to other platforms. Thus, the model is primarily applicable to social media environments with similar linguistic and engagement patterns. Second, the limited dataset size may restrict the model’s ability to learn more complex patterns in language use and emoji behavior.

Further work would explore larger and more diverse datasets across additional platforms, as well as alternatives emoji representations and additional engagement metrics such as comments or shares to improve the robustness of engagement prediction models.

## References

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